William Powell

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Data Scientist with an MSc in Applied Machine Learning from Imperial College London, experienced in developing and deploying ML models, building agentic AI systems, and solving real-world challenges at scale. Currently evaluating conversational AI that serve several million customers in the consumer lending space.

Education

Imperial College London October 2023 - September 2024
Applied Machine Learning MSc *Top of cohort; Distinction, Dean's List 2024*

University of Bath September 2019 – June 2023
Integrated Mechanical and Electrical Engineering BEng (Hons) *Top of cohort; First Class Honours*

Experience

Senior Data Scientist at Lendable February 2025 – Present

- Built an agentic, conversational AI responding to over 1 million interactions annually; optimized via evals for both quality and resolution.
- Built AI document parsers for income and identity verification that handle over 250k documents per year using LLM fine tuning and GEPA.
- Trained and deployed a deep learning model for pricing unsecured consumer loans, adding an additional £3m in originations per year.

Machine Learning Research Intern at Mixedbread.ai October 2024 – January 2025

- Helped to research, implement, and optimize state-of-the-art retrieval models.
- Integrated multi-modal AI schema extraction into an upcoming product release.

Research Engineer at BrainPatch.ai June 2021 – September 2024

- Designed from conception to deployment the machine learning pipeline, software, firmware, and hardware to wirelessly stream EEG/ECG data to the cloud. Implemented model inference via cloud computing to process data and transmit to a web interface in real time.
- Device used in trials to remotely monitor the biosensing information of the participants in real time. Now integrating it into a commercial machine learning consumer product to detect stress.

Software Engineer at AB Dynamics September 2021 – August 2022

- Led embedded software development for a new product, integrating CAN protocol with STM32 microcontroller peripherals using C++. Device developed and launched to customers in six months.
- Researched and prototyped a neurostimulation device to mimic motion in a static driving simulator.

Founder of PhoneCave November 2019 – September 2020

- Created a start-up, PhoneCave, that aims to get people off their phones.
- Designed and manufactured the electronics, firmware and hardware. Launched and shipped PhoneCaves to over 10 countries worldwide.

Publications & Research

Modular Sliding Cross-Attention Network for Early-Onset Real-time Stress Detection.
(Forthcoming)

- As an extension to my Master's thesis, we propose a novel architecture and attention mechanism inspired by the Transformer, commonly used in NLP tasks. This adaptation is designed for real-time biosignal classification.
- The architecture required significantly smaller feature extraction windows, resulting in one fifth of the compute to achieve state-of-the-art accuracies in stress classification datasets.

Sliding Cross-Attention Network for Low-Latency Wearable Motion Capture on Edge Devices.
(Forthcoming)

- From my previous paper it quickly came apparent that this architecture is well suited not only for biosignal classification but other real-time, compute limited classification tasks.
- The idea is to detect human motion using wearable sensors instead of external cameras with the eventual goal of capturing human motion in users' own homes which will be utilised to diagnose various patient conditions.

Rethinking Stress Monitoring: Convenient Modular Early-Onset Multimodal Stress Detection with Attention Score Caching. *(MSc Thesis, Imperial College London)* Paper available here.

- Developed a modular system for real-time stress detection using multimodal data and optimized attention mechanisms.

Design and Implementation of a Single-Lead Chest Strap ECG Recorder for Stress Classification. *(BEng Dissertation, University of Bath)* Paper available here.

- Engineered a custom hardware solution and lightweight ML methods for stress classification via ECG.

Projects

Contribution to Open Source Machine Learning Accelerator Framework Code Available Here

- Integrated TensorRT and ONNX Runtime into a joint Imperial-Cambridge research codebase to enable post-training quantization (PTQ). This project aims to future-proof hardware accelerators by ensuring compatibility with rapidly evolving ML models.

Design of a Self-Organising Multi-Agent System Code Available Here

- Co-lead the infrastructure team to design a base platform architecture for 70+ students to utilize. Six publications written at Imperial using this system environment.

COVID Volunteering - 3d Printing PPE for National Health Service

- Collaborated in a 3D printing community to print PPE for NHS staff, enabling the creation of approximately 20,000 additional face shields due to slicer optimizations.

Technical Skills

Languages: English (native), Russian (B1), French (B1), Italian (A1)
Programming Languages: Python, SQL, C++, C, Go, React, MATLAB, Julia, JS, Bash
ML Tools: PyTorch, TensorFlow, Keras, OpenCV, CUDA, DSPy
Developer Tools: Git, Linux, Docker & Kubernetes, FastAPI, PostgreSQL, AWS, GCP

Activities and Interests

- Photography, running, sailing, skiing, freediving, investing, philosophy.